

XENIX[®] System V/286 Operating System

Installation Guide

XENIX® System V/286

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XENIX Installation Guide

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Chapter 1

Introduction

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The XENIX® operating system is a powerful system of programs that gives your computer the same multiuser, multitasking capabilities as many large and expensive computers. This guide explains how to install XENIX System V on your new Intel 286(80286)-based computer.

1.1 XENIX Software Systems

The complete XENIX operating system is a set of three software systems:

- The Timesharing System (includes the Help Page System)
- The Software Development System
- The Text Processing System

The Timesharing System, herein referred to as the Base System, contains the XENIX programs you need to create user accounts, manage file systems, and perform system maintenance tasks. The Help Page System provides on-line manual pages for XENIX programs and utilities.

The Software Development System contains the XENIX programs you need to create, compile, link, and debug assembly and high-level language programs.

The Text Processing System contains the XENIX programs you need to create, edit, and typeset documents.

Each package is installed separately. Installation of the Base System is required before you can install the Help Page System, Text Processing, or Software Development Systems.

1.2 What You Need

To install the Timesharing System on your computer you need

- One fixed disk with a minimum of 15 megabytes of storage
- One high-capacity floppy disk drive

If the fixed disk and floppy disk drive are not already connected to your computer, you must connect them according to your computer's operations manual before continuing with the XENIX system installation. If you intend to install the Text Processing and/or the Software Development Systems, you may need additional fixed disk space. The complete XENIX system requires at least 15 megabytes of storage.

1.3 Before you begin...

Before you begin installing XENIX, set up your computer as described in your computer's operations manual. Read about turning on your computer and fixed disk (also called "hard disk"), the location of the primary floppy diskette drive, how to insert floppy diskettes, and how to reset your computer.

Be sure to type names, numbers, and spaces exactly as they are shown, then press ENTER at the end of each line. You can use the BACKSPACE key (or CTRL-H) to correct any typing errors before pressing ENTER. To type a CTRL-H, hold down the CTRL key while typing 'H'. To delete an entire line before pressing ENTER, press the DELETE key (or CTRL-U). Your computer's manual describes the key labels on your computer for ENTER, BACKSPACE, DELETE, and CTRL.

Some manuals refer to the ENTER key as RETURN.

Chapter 2

Installing XENIX

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The following sections detail each step in the installation procedure. Follow each step in the order it is presented. When you have completed the installation, keep this guide and the distribution diskettes in a safe place. You might need them again if you wish to reinstall the system for any reason.

2.1 Formatting the Fixed Disk

Before you can install XENIX on your fixed disk, you must format it according to the instructions in your computer's operations manual.

If you are using a nonstandard fixed disk, one whose drive type is not in the drive type table, you may want to specify the number of heads and cylinders during the format procedure. Doing so ensures that your drive is used to its maximum capabilities. For more information on the drive type table, see **hd (M)** in the *Reference Manual Volume I*.

To change the number of heads and cylinders on a Tandy® 3000, follow the instructions in your computer's operations manual for formatting the fixed disk. At the prompt, "Is this correct? (Y/N)," answer 'N'. You are then prompted to enter the number of heads, cylinders, and interleave factor for the fixed disk. Enter the correct parameters for your fixed disk, then continue the formatting procedure as described in your computer's operations manual. XENIX performs best at an interleave of 4 to 1.

When you have finished formatting the fixed disk, you may proceed to the next step of the installation procedure.

2.2 Starting XENIX From Floppy Diskette

You must first start up your computer from the distribution diskette labeled "Installation." To do so, follow these steps:

1. Turn on your computer and all peripherals. Follow the instructions given in your computer's operations manual.
2. Insert the Installation diskette into the primary floppy diskette drive. Your computer loads and runs the XENIX *bootstrap* program from the diskette. The screen shows:

```

SYSV    XENIX Boot

Enter:   hd program
         fd program
         dos
         cf [-c conf_file] [device program]

Press Enter for default:   fd /xenix.fd
:
```

Do not press ENTER. If you do not see this prompt, remove the diskette, turn off your computer and try again.

Proceed to the next step: Creating the Bad Track Table.

2.3 Creating the Bad Track Table

The Bad Track Table tells XENIX which tracks on the fixed disk are bad and should not be used. It is normal for a fixed disk to have a few unusable tracks.

Use the **badtrack** program on the Installation diskette to create the badtrack table. The **badtrack** program scans your fixed disk for bad tracks and lists those it finds. The program then creates the bad track table on the fixed disk.

To create the bad track table, follow these steps:

1. At the "XENIX" Boot" prompt, type:

```
fd /etc/badtrack
```

2. Press ENTER. The screen shows:

```
Loading
```

3. After the program is loaded, the screen shows:

```
Loaded, press Enter to start
```

4. Press ENTER. The **badtrack** program scans the fixed disk for bad tracks. **Badtrack** displays:

```
drive 0 has xxx cylinders and x heads  
scanning drive 0  
cylinder n
```

where *n* is the current cylinder number.

5. When **badtrack** has completed its scan of the disk, it displays:

```
Map for drive 0 is:
```

followed by a list of all bad tracks found. Now, **badtrack** asks if you want to add any additional bad tracks. A **Media Error Map** may have come with your fixed disk.

If it shows any additional bad tracks, enter them at this time. **This is CRITICAL to the operation of XENIX.** For further information on the **badtrack** utility, see the Commands (M) section of the *Reference Manual Volume I*.

Note: If you have a second fixed disk attached to your system, **badtrack** asks if you wish to check the secondary fixed disk. If you enter 'Y', **badtrack** continues by checking the secondary drive.

When **badtrack** is complete, XENIX returns to its boot prompt:

```

SYSV   XENIX Boot

Enter:   hd program
         fd program
         dos
         cf [-c conf_file] [device program]

Press Enter for default:   fd /xenix.fd
:
```

Proceed to the next step: Starting the XENIX Installation.

2.4 Starting the XENIX Installation

You are now ready to start up XENIX from the Installation diskette. Follow these steps:

1. At the "XENIX Boot" prompt, press ENTER. The screen shows:

```

fd /xenix.fd
Loading
```

2. This loads the XENIX system from the Installation diskette. The screen shows:

```

Loaded, Press Enter to start
```

3. Press ENTER. XENIX displays its startup information and the prompt:

```

No single-user login present
Entering System Maintenance Mode
#
```

Proceed to the next step: Initializing Your Fixed Disk.

2.5 Initializing Your Fixed Disk

The fixed disk initialization program, **hdinit**, creates a partition table for your fixed disk. It also copies important XENIX files to the fixed disk and prepares the fixed disk to receive other XENIX programs and files.

To run **hdinit**, follow these steps:

1. At the # prompt, type:

```
hdinit
```

2. Press ENTER. The **hdinit** program loads and displays introductory information about the installation process, followed by:

```
Press ENTER to continue or DEL to abort the installation:
```

3. If you have not backed up your files and wish to do so, press DEL and back up all files you wish to save before continuing with the installation. Otherwise, if all of your files have been backed up, press ENTER. When you press ENTER, the screen displays information about choosing swap area size and the prompt:

```
Do you want to use the default swap area size? (y,n)
```

4. The default swap area size is adequate for most purposes, especially if you have installed 3 megabytes or less of memory. See Appendix B for more information about swap area size if you have installed more than 3 megabytes of memory.

Type 'Y' and press ENTER if you wish to use the default swap area size of 3000 Kbytes. Then, proceed to Step 5. Otherwise, if you wish to use a larger swap area size, type 'N' and press ENTER. The **hdinit** program displays:

```
Enter swap size in Kb (range 3000-8000):
```

The **hdinit** program calculates the largest possible swap area size available on the fixed disk assuming that 6 megabytes will be allocated to the root file system and that the minimum 6 megabytes will be allocated to the user file system. In this case, it calculated 8000 blocks were available for swap area. Enter the swap area size that suits your needs and press ENTER. Remember that as the swap area size grows, the amount

of space available on the fixed disk for the user partition decreases.

5. The **hdinit** program displays the information about the options available for partitioning your fixed disk followed by:

If you choose option 1 or 2, all current files on the fixed disk will be destroyed. You must back up all user files you wish to save before continuing to install XENIX on the fixed disk.

Enter option number:

6. Choose the option that suits your needs:

- Option 1 if XENIX is going to be the only operating system on your fixed disk.
- Option 2 if you want to reserve space for another operating system (such as MS-DOS) in addition to XENIX. If you choose this option, you will be asked how much space you wish to reserve for this other operating system. The valid range of values is 1 to 4500, which represents the number of 1K byte blocks you wish to reserve. When the program poses this question, type in the number and press the ENTER key.
- Option 3 if you wish to add XENIX partitions to the current partition table on the fixed disk. This option lets you install XENIX on the fixed disk starting somewhere other than the beginning of the disk. This option also allows other operating systems, such as DOS, to exist on the fixed disk along with XENIX. If you choose option 3 skip to the section "Nonstandard Installation."

7. Select the appropriate option, type the corresponding option number, and press the ENTER key.

If you selected option 3, go to "Nonstandard Installation." If you selected option 1 or 2 proceed with these instructions. When you run **hdinit** it creates the *disk partitions* for a new XENIX system. A disk partition is an area on your fixed disk that has been reserved for a specific use. The **hdinit** program displays:

The root partition is now being created.

The **hdinit** program creates the appropriate partitions and continues with the installation procedure.

Installing XENIX

8. After the disk partitions are created, **hdinit** asks for your time zone. It displays the following message:

Enter the name of your local time zone by choosing any one of the following time zones or create a custom time zone by choosing Option 7 (Other):

1. Greenwich
2. Atlantic
3. Eastern
4. Central
5. Mountain
6. Pacific
7. Other

Enter option number:

9. Select the appropriate time zone, type the corresponding option number, and press the ENTER key. If you select option 7, **hdinit** prompts you for additional information.
10. The **hdinit** program displays several messages about what it is doing. When the program is finished, it displays the following message and halts the system:

Now do the following things:

1. Wait for the shutdown message.
2. Remove the Installation Diskette.
3. Press ENTER to reboot XENIX from the fixed disk.
4. Follow the instructions of the automated installation process.

****Normal System Shutdown****

Press ENTER to reboot:

11. Remove the Installation floppy disk from the floppy disk drive and press ENTER as instructed. This loads the XENIX boot program from the fixed disk. XENIX displays its boot prompt.

Note: XENIX automatically boots if you do not press ENTER within a few

minutes. If this should accidentally happen, XENIX shuts itself down. Press ENTER to reboot and try again. Automatic booting does not work at this stage of the installation process. You are now ready to start the XENIX system from the fixed disk. At the boot prompt, press ENTER. The program displays the message:

```
:
hd /xenix
Loading
```

XENIX loads from the fixed disk and displays its startup information. This information includes the status of optional equipment such as disk cartridge (cd), tape cartridge (mt), serial adapter (sa), and multi-channel board (mc). Then, the following prompt is displayed:

```
No single-user login present
Entering System Maintenance Mode
```

Proceed to the next step: Installing XENIX Programs and Files.

2.6 Installing XENIX Programs and Files

You are now ready to install the XENIX programs and files from the distribution diskettes to your fixed disk. XENIX displays information on installing the XENIX programs. XENIX then prompts:

```
Press ENTER to continue:
```

Before proceeding any further, be sure you have all the diskettes for the base XENIX system near you and in the correct numerical order.

When you are ready to continue, press ENTER and follow these steps:

1. XENIX automatically executes `/etc/firsttime`, which executes **xinstall** for the Base System.
2. After **xinstall** does an initial check for the file system, it displays some information on inserting diskettes followed by the prompt:

```
First diskette? [y,n]
```

3. Insert the first distribution diskette into the primary floppy drive. Type 'Y' and press ENTER.
4. The **xinstall** program displays the names of the files as it copies them from the distribution diskette to the fixed disk.

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5. After all files are copied to the fixed disk, **xinstall** displays:

```
Next diskette? [y,n]
```

or

```
tar: please mount new volume
```

6. Remove the diskette from the primary floppy drive, and insert the next diskette from the distribution set. Be sure that you insert the diskettes in the correct order.
7. If the message “Next diskette” is displayed, type ‘Y’ and press ENTER. Otherwise, simply press ENTER.
8. Continue inserting the diskettes when prompted until there are no more diskettes to be copied to the fixed disk. Then, type ‘N’ and press ENTER at the “Next diskette” prompt. The **xinstall** program then sets the correct default permissions for the files it just installed.
9. XENIX now displays:

```
XENIX System V/286 BASE SYSTEM INSTALLATION COMPLETE  
Do you wish to install the Software Development  
Package now (y/n)?
```

If you purchased the Software Development System and wish to install it at this time, type ‘Y’ and press ENTER. XENIX automatically executes the **xinstall** program for installing the Software Development System. When prompted, insert the diskettes (in order) as described earlier. When there are no more diskettes to be copied, type ‘N’ and press ENTER at the “Next Diskette” prompt.

If you do not want to install the package, type ‘N’ and press ENTER.

10. XENIX now prompts:

```
XENIX System V/286 SOFTWARE DEVELOPMENT PACKAGE  
INSTALLATION COMPLETE  
Do you wish to install the Text Processing  
Package now (y/n)?
```

If you purchased the Text Processing System and wish to install it at this time, type ‘Y’ and press ENTER. XENIX automatically executes the **xinstall** program for installing the Text Processing System. When prompted, insert the diskette as described earlier. After all files have been copied, type ‘N’ and press ENTER at the “Next Diskette” prompt.

If you do not want to install the package, type 'N' and press ENTER.

11. When the installation is complete, **xinstall** displays the message:

```
XENIX SYSTEM V/286 TEXT PROCESSING PACKAGE
INSTALLATION COMPLETE
Do you wish to install the On-Line Help Pages
Package now (y/n)?
```

If you wish to install the on-line manual pages that came with your base system, you can do so at this time. This involves the creation of a keyword file that takes about 25 minutes to create. If you wish to install it at this time, type 'Y' and press ENTER. XENIX automatically executes the **xinstall** program for installing the Help Pages. When prompted, insert the diskettes (in order) as described earlier. When there are no more diskettes to be copied, type 'N' and press ENTER at the "Next Diskette" prompt.

If you do not want to install the package, type 'N' and press ENTER.

12. XENIX now displays some information and proceeds to shut down the system.
13. When XENIX displays the XENIX Boot prompt, press ENTER. The display shows:

```
** Normal System Shutdown **
```

```
Press Enter to reboot:
```

14. Press ENTER to reboot the system. XENIX starts up and displays its Boot prompt. Press ENTER. XENIX displays its startup information and the following prompt:

```
Type Ctrl-d to proceed with normal startup,
(or give root password for system maintenance):
```

15. Type CTRL-D by holding down the CTRL key while pressing 'D'. If you have any problems, refer to the explanation of normal startup in the *XENIX Operations Guide*.

Your XENIX system is now completely installed on your fixed disk.

Proceed to the next step: Creating the Super-User Password.

2.7 Creating the Super-User Password

The super-user password protects the system from unauthorized use. It is important that you create a super-user password immediately after the system is installed to ensure maximum protection and prevent unnecessary use of the super-user account. For a complete description of super-user, see the *XENIX Operations Guide*.

To create the super-user password, follow these steps:

1. At the “Login” prompt, type:

```
root
```

2. Press ENTER. At the # prompt, type:

```
passwd root
```

3. Press ENTER. XENIX prompts:

```
New password:
```

4. The password can be any sequence of letters, numbers, and/or punctuation marks. You should use a minimum of 5 characters. Choose something that is not easily guessed, but that you can remember. If you forget the root password, you must reinstall XENIX. For more information on passwd, refer to passwd (C).
5. Type the new password and press ENTER. As a safety measure, XENIX does not display the password as you type it.
6. XENIX prompts:

```
Re-enter new password:
```

7. Retype the password and press ENTER. This step ensures that you did not mistype the password. Both entries must match before XENIX changes the password.

Proceed to the next step: Creating the First User Account.

2.8 Creating the First User Account

Next, you create the system's first user account, "demo." This demonstration account is a temporary workspace on the system you can use to practice with the XENIX system. Later, after installation is complete and you are familiar with the XENIX commands, you can remove the account and create private accounts for all the users on the system.

To create the first user account, follow these steps:

1. At the # prompt, type:

```
mkuser
```

and press ENTER. The screen shows:

```
Newuser
```

```
Add a user to the system
```

```
Do you require detailed instructions? (y|n|q):
```

2. For now, type 'N' and press ENTER. You can read the instructions at another time. The screen displays the following prompt:

```
Enter new user's login name:
```

3. Because we wish to name the first account "demo," type:

```
demo
```

and press ENTER.

4. The **mkuser** prompt for a group name appears:

```
Do you want to use the default group? (y|n|q):
```

5. For now, type 'Y' and press ENTER. The following prompt appears:

```
Please enter at least 5 characters for the
password
Enter password:
```

6. Type:

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password

and press ENTER. You are prompted to retype the password as a safety precaution. Next, **mkuser** prompts you for a shell type:

ENTER Shell type (1, 2, 3 or 4) and press ENTER

7. The shells available are: 1) Standard (Bourne) Shell, 2) Visual Shell, 3) Cshell, and 4) /usr/lib/uucp/uucico. Enter the number that corresponds to the shell you want to use.
8. The screen displays a prompt for comments:

```
Please Enter Comment: >-----  
>
```

9. Type "demo account" and press ENTER.
10. The program then displays information about the account and asks if you wish to change anything. Type 'N' and press ENTER. **Mkuser** asks if you wish to create another account.
11. Type 'N' and press ENTER.

You have now created your system's first user account. Later, when you turn to the *XENIX User's Guide*, you can use this account to practice logging in, making directories, and running programs.

Proceed to the next step: Installation Complete.

2.9 Installation Complete

Your XENIX installation is now complete.

If you are familiar with the XENIX Operating System, you can continue with normal startup and begin working. Simply hold down your computer's CTRL key, and type the letter 'D'. Refer to the explanation of normal startup in the *XENIX Operations Guide* if you have problems.

If you are not familiar with the XENIX Operating System, we recommend that you halt the system. Turn to the *XENIX User's Guide* and the *XENIX Operations Guide* to learn how to start the system, how to log in, and how to run programs.

To halt the system you must be logged in as root. Follow these steps:

1. At the # prompt, type:

```
sync;/etc/haltsys
```

and press ENTER. The system displays the following message:

```
** Normal System Shutdown **
```

```
Press Enter to reboot:
```

2. Turn off the power to the computer. Always be sure that you see the shutdown message before powering down the computer.





Chapter 3

Installing Optional Software

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 - 3.4 Installing Help Pages 3-4
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The following sections describe how to install the Software Development System, the Text Processing System, and the On-Line Help Page System if you did not install them during the installation procedure.

3.1 Before You Begin...

Before you begin, be sure you have all the appropriate distribution diskettes nearby and in the proper numerical order.

You must log in as super-user in system maintenance mode to install the optional packages. If you just completed the installation, you are already in system maintenance mode. Otherwise, you must:

- Halt the system using `/etc/haltsys`. Remember to notify other users that you are shutting down the system.
- Start up the system normally by pressing ENTER at the XENIX Boot prompt. Enter system maintenance mode by giving the root password at the prompt:

Type Ctrl-d to proceed with normal startup,
(or give root password for System Maintenance):

You can find more information on shutting down the system and entering system maintenance mode in the *XENIX Operations Guide*.

3.2 Installing the Software Development System

1. At the # prompt, in system maintenance mode, type:

```
xinstall soft
```

and press ENTER.

2. The **xinstall** program displays some information as it checks the *usr* file system and then prompts:

```
First Diskette? [y,n]
```

3. Insert Diskette # 1 of the Software Development System distribution set into the primary floppy diskette drive.
4. Type 'Y' and press ENTER. The **xinstall** program copies the files from the floppy diskette to the fixed disk. After all files are copied, the screen shows:

```
Next Diskette? [y,n]
```

5. Remove the diskette from the floppy drive and insert the next diskette. Type 'Y' and press ENTER.
6. *Continue inserting diskettes, when prompted, until all diskettes are copied to the fixed disk.*
7. When all the diskettes are copied, type 'N' and press ENTER at the "Next Diskette?" prompt.

When the installation is complete, the screen shows:

```
Installation complete
```

To install the kernel configuration software (Config Disk), type:

```
xinstall config
```

and press RETURN. Follow Steps 2 - 7 above for inserting diskettes. The installation is complete when the system displays "Installation complete."

3.3 Installing Text Processing

1. At the # prompt, in system maintenance mode, type:

```
xinstall text
```

and press ENTER

2. The **xinstall** program displays some information as it checks the *usr* file system and then prompts:

```
First Diskette? [y,n]
```

3. Insert the Text Processing distribution diskette into the primary floppy diskette drive.
4. Type 'Y' and press ENTER. The **xinstall** program copies the files from the floppy diskette to the fixed disk. After all files are copied, the screen shows:

```
Next Diskette? [y,n]
```

5. Type 'N' and press ENTER.

When the installation is complete, the screen shows:

```
Installation complete
```

3.4 Installing Help Pages

1. At the # prompt, in system maintenance mode, type:

```
xinstall help
```

and press ENTER

2. The **xinstall** program displays some information as it checks the *usr* file system and then prompts:

```
First Diskette? [y,n]
```

3. Insert Diskette # 1 of the Help Pages distribution diskette into the primary floppy diskette drive.
4. Type 'Y' and press ENTER. The **xinstall** program copies the files from the floppy diskette to the fixed disk. After all files are copied, the screen shows:

```
Next Diskette? [y,n]
```

5. Remove the diskette from the floppy drive and insert the next diskette. Type 'Y' and press ENTER.
6. When the screen shows the "Next Diskette" prompt again, type 'N' and press ENTER.

When the installation is complete, the screen shows:

```
Installation complete
```



Chapter 4

Adding a Secondary Fixed Disk

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If your system is heavily used, you may want to add a secondary fixed disk to your XENIX system. There are five steps to follow when adding a secondary fixed disk. They are:

1. Connecting the secondary fixed disk to your computer
2. Formatting the secondary fixed disk for use with XENIX
3. Preparing the secondary fixed disk for use with XENIX
4. Partitioning the secondary fixed disk
5. Creating file systems on the secondary fixed disk partitions

4.1 Connecting the Secondary Fixed Disk to Your Computer

Before you connect the secondary fixed disk to your computer, you must shut-down your XENIX system. See the section titled “Installation Complete” for information on shutting down your system. When XENIX displays the message:

```
** Normal System Shutdown **
```

```
Press Enter to reboot:
```

turn off the computer and connect your secondary fixed drive as explained in the secondary fixed disk’s owner’s manual.

Note: Your manufacturer may have provided diagnostics along with the fixed disk. Be sure that your fixed disk passes these diagnostics before you connect it to your XENIX system.

After you have connected the secondary fixed disk, you must update the configuration information of your computer. See the information on the “setup” utility in your computer’s operations manual.

4.2 Formatting the Secondary Fixed Disk

Before you can use your secondary fixed disk with XENIX, you must format it according to the instructions in your computer’s operations manual.

If you are using a nonstandard fixed disk, one in which the drive type is not in the drive type table, you may want to specify the number of heads and cylinders during the format procedure. Doing so ensures that your drive is used to its maximum capabilities. For more information on the drive type table, see `hd (M)` of the *Reference Manual Volume I*.

To change the number of heads and cylinders on a Tandy 3000, follow the in-

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structions in your computer's operations manual for formatting the fixed disk. At the prompt, "Is this correct? (Y/N)", answer 'N'. You are then prompted to enter the correct parameters for your fixed disk, then continue the formatting procedure as described in your computer's operations manual.

4.3 Preparing the Secondary Fixed Disk for Use with XENIX

To format your secondary fixed disk follow these steps:

1. Turn on your computer and insert the XENIX Installation Diskette into the primary floppy diskette drive. The screen shows:

```
SYSV    XENIX Boot

Enter:   hd program
         fd program
         dos
         cf [ -c conf_file ] [ device program ]

Press Enter for default:      fd /xenix.fd
:
```

type:

```
fd /etc/badtrack2
```

and press ENTER.

2. Refer to the **badtrack (M)** command and create a bad track map for the secondary fixed disk. A **Media Error Map** may have come with your secondary fixed disk. Be sure to enter any additional bad track that it notes when you are asked to enter additional bad tracks.
3. When the badtrack utility has completed, XENIX displays:

```
SYSV    XENIX Boot

Enter:   hd program
         fd program
         dos
         cf [ -c conf_file ] [ device program ]

Press Enter for default:      fd /xenix.fd
:
```

4. Type:

```
hd /xenix
```

5. Press ENTER to boot off the fixed disk. Type CTRL-D to begin normal startup, log in as root, and proceed to the next step.

4.4 Partitioning the Secondary Fixed Disk

In order for XENIX to take advantage of storage space available on your secondary fixed disk, you need to know some of its specifications. To determine this information, follow these steps:

1. Find the number of Cylinders, Heads, and Sectors/Track as described in your secondary fixed disk's owner's manual.
2. Calculate the number of BLOCKS on the secondary fixed disk by using the following formula:

$$\text{Blocks} = (\text{Cylinders} * \text{Heads} * \text{Sectors} / 2) - 100$$

To calculate the number of cylinders, use this formula:

$$\text{Cylinders} = (K\text{-bytes} + 9) / (\text{Heads} * \text{Sectors} / 2)$$

For example, if you have a standard "Type 2" fixed disk, your manual indicates that the fixed disk has 615 Cylinders, 4 Heads, and 17 Sectors/Track. Therefore, for a "Type 2" fixed disk, the number of blocks would be 20810 or:

$$\text{Blocks} = (615 * 4 * 17 / 2) - 100$$

3. You must now decide how to allocate these blocks on your secondary fixed disk. You can create one XENIX user partition, or you can create several different partitions. You can specify that each partition is to be a XENIX user partition, or another type of operating system partition, such as DOS.

For example, if you wish to create one XENIX user partition that occupies the entire fixed disk, you would allocate all blocks to that partition.

If you wish to create one XENIX user partition and one DOS partition, you will need to decide how many blocks to allocate to the XENIX partition and how many to allocate to the DOS partition. You may have up to 3 partitions on the fixed disk (not including the bad track table, which must always be present for XENIX).

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Note: You cannot create non-XENIX partitions while running XENIX. To create a DOS partition, please refer to your DOS manual.

Once you know the specifications of your fixed disk, you are ready to perform the actual partitioning. Type:

```
/etc/fdisk /dev/rhd10
```

Press ENTER. The **fdisk** utility lets you divide the secondary disk (/dev/rhd10) into logical partitions. To display the current partition information, type, at the * prompt, 'p' and press ENTER. The **fdisk** program displays the partition map of the disk. It should look something like this:

Partition	Type	Active	Start Cyl	End Cyl	Sectors
1	BTT (255)	N	0	0	1
2					
3					
4					

This table shows the Bad Track Table (BTT), created by /etc/badtrack, on Partition 1. Partitions 2-4 are available for your use. Suppose you wish to create one XENIX user partition and one DOS partition on the secondary fixed disk. At the "*" prompt, type 'c' and press ENTER. The **fdisk** program prompts:

```
Enter partition number [1-4]:
```

Because the Bad Track Table occupies Partition 1, we chose Partition 2 for the XENIX partition. Type '2' and press ENTER. The **fdisk** program asks:

```
Enter starting cylinder number [0-614]:
```

You can specify Cylinder 0 as the starting cylinder of another partition, as **fdisk** will not overwrite the Bad Track Table. Type 0 and press ENTER. Now, **fdisk** displays the following prompt:

```
Enter length in 1K blocks:
```

Earlier you calculated that there were 20,810 blocks available for the partitions. For this example, give the XENIX partition 10,810 blocks and the DOS partition 10,000 blocks. Type '10810' and press ENTER. The **fdisk** program prompts:

```
Enter type (2=system, 3=user):
```

The type of partition will always be Type 3 to indicate a XENIX user partition.

Type '3' and press ENTER.

You are now ready to save the new partition map to the fixed disk. To do so, type 'w' and press ENTER. To quit the **fdisk** program, type 'q' and press ENTER.

If you wish to devote the entire fixed disk to a XENIX user partition, follow the example above. However, when **fdisk** asks:

Enter length in 1K blocks:

Respond with the total number of blocks available and press ENTER. For this example, you would enter 20810.

If you wish to use a different partitioning scheme, use the above examples as a guide and consult the documentation for **fdisk** (M) in the *Reference Manual Volume I*.

4.5 Creating File Systems on the Secondary Fixed Disk Partition

In order for your XENIX system to recognize the secondary fixed disk partitions, you must create file systems on the partitions.

Before you can create file systems, you must create a link between the special device file that actually represents the partition and a symbolic device name you want to call the partition. This is done with the **ln** (C) command. (Special device files that actually represent partitioned areas of a disk are defined in **hd** (M) of the *Reference Manual Volume I*.)

Using the previous example, the new XENIX partition was created as Partition 2 on the secondary fixed disk. This is represented by the special device name **/dev/hd12**. The "1" represents the device name, and the "2" represents the partition number.

The file system, once created, needs to be mounted onto a directory. For this example, use the directory name **/usr2**. This is done with the **mkdir** (C) and **mount** (C) commands.

Follow these steps:

1. Link the block device by typing the following command and pressing ENTER:

```
ln /dev/hd12 /dev/usr2
```

2. Link the character device by typing the following command and press-

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ing

ENTER:

```
ln /dev/rhd12 /dev/rusr2
```

3. Make the intended mount directory by typing the following command and pressing ENTER:

```
mkdir /usr2
```

If you have made more than one XENIX partition, use the same method to create devices, names, and directories for each partition. You do not have to tell XENIX about any DOS partitions you've created. No special commands are needed for those partitions.

You are now ready to create a file system for each XENIX partition. To do so, use the **makefs** (C) command.

Type:

```
makefs
```

and press ENTER. A menu appears asking on which type of disk you wish to make a file system. Select Option 1, a fixed disk. Next, the screen shows:

```
On which hard drive (0-1)
```

Enter a "1" to specify the second fixed disk. The **makefs** program displays the partition table of your secondary fixed disk which was just created and prompts:

```
On which XENIX partition (1-4 or <DEL> to abort)?
```

Enter the number of the partition on which you wish to make a file system. If you have more than one XENIX partition, choose one of them and then run this command again for the others. If you have only one XENIX partition, enter the number listed beside it.

For example, if you made one XENIX partition to fill the entire secondary fixed disk, it will be on Partition 2, so enter "2." If you press DEL, **makefs** returns you to the original menu. You can then start again or quit. If you enter a valid XENIX partition number, **makefs** displays a message giving the drive number, partition, and size of the file system about to be created. Next, the screen shows:

Press <RETURN> to continue or to abort:

If you press ENTER, **makefs** does not create the file system, but displays:

When you press ENTER, **makefs** returns you to the original menu when you can make another file system on a different XENIX partition or you can quit.

Each partition on which you create a file system is available for you to use as a mountable file system. To mount the example partition file system onto the /usr2 directory, type:

```
/etc/mount /dev/usr2 /usr2
```

and press ENTER. The partition file system is now part of the XENIX file structure and will stay that way until you unmount it. To unmount the example partition file system, type:

```
/etc/umount /dev/usr2
```

and press ENTER. For further information refer to the **mount (C)** and **umount (C)** command in the *Reference Manual Volume I*.

4.6 Automatically Mounting Partitions on the Secondary Fixed Disk

This section shows a sample /etc/rc.user shell file that you can create to have XENIX automatically mount XENIX partitions from the secondary fixed disk during startup. It assumes that you are familiar with shell commands and a XENIX editor. For further information on shell commands, refer to **sh (C)** in the *Reference Manual Volume I*.

When XENIX begins its normal startup, it performs many functions before it allows any user to log into the system. One function XENIX performs is mounting the partition file system '/dev/usr' onto the '/usr' directory. You can inform XENIX of other partition file systems for it to mount during the startup procedure via the /etc/rc.user file.

During startup, the /etc/rc file checks for the existence of a file called "/etc/rc.user". If this file exists, then /etc/rc automatically executes it as a shell file.

The following example /etc/rc.user tells XENIX to mount the partitioned file systems /dev/usr2 and /dev/usr/3. They would remain mounted until you unmount them.

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4.6.1 Example

```
# This is an example of the /etc/rc.user file. All
# shell commands in the file are executed before
# XENIX allows any users to log in.

/etc/mount /dev/usr2 /usr2      # Mount file system
if test S?=2                    # /dev/usr2
then                             # If XENIX couldn't mount
    echo "Cleaning /dev/usr2"   # it, then clean it using
    fsck -y -t /tmp/fsck$$/dev/usr2 # the fsck(C) command, and
    /etc/mount /dev/usr2 /usr2   # try to mount it again.
fi >/dev/console 2>&1

/etc/mount /dev/usr3 /usr3      # Mount file system
if test $?=2                    # /dev/usr3
then                             # If XENIX couldn't mount
    echo "Cleaning /dev/usr3"   # it, then clean it using
    fsck -y -t /tmp/fsck$$/dev/usr3 # the fsck(C) command, and
    /etc/mount /dev/usr3 /usr3   # try to mount it again.
fi >/dev/console 2>&1
```


Appendix A

Nonstandard Installations

You can modify the standard installation procedure of disk partitions by selecting Option 3 in Step 3 of “Initializing Your Fixed Disk.” This option lets you create and define your own partition size and location. Use this option if another operating system, like DOS, is already installed on the fixed disk, and you want to add XENIX. If DOS or another operating system is already installed on the fixed disk and you wish to install XENIX, be sure:

- There is a 9000K-byte contiguous area on your fixed disk for the *root* file system.
- There is at least a 6000K-byte contiguous area on your fixed disk for the *user* file system.

Note: To calculate how many cylinders will be used for your partition, use the following formula:

$$\text{Cylinders} = (K\text{-bytes} + 9) / (\text{Heads} * \text{Sectors} / 2)$$

To calculate the number of blocks, use this formula:

$$\text{Blocks} = (\text{Cylinders} * \text{Heads} * \text{Sectors} / 2) - 100$$

Your operations manual will have the information on the number of heads and sectors for your fixed disk.

1. Select Option 3 by typing ‘3’ and pressing ENTER.
2. The **hdinit** program displays a table showing the size and type of the partitions already existing on the fixed disk. Next, it displays the prompt:

Enter the root partition number [1-n]:

3. Be sure there are 2 empty or not defined partitions available unless you are reinstalling XENIX. In this case, you can select the XENIX partition. One partition is needed for the XENIX *root* file system and the other for the XENIX *user* file system.
4. Choose a partition number for the XENIX *root* file system. Type the number and press ENTER.
5. The screen displays the prompt:

Enter the user partition number [1-n]:

6. Choose a partition number for the XENIX *user* file system. Type the

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number and press ENTER.

7. The screen shows:

```
Enter the starting root cylinder number [0-n]:
```

Type the starting cylinder number for the *root* partition and press ENTER. The *root* partition must have at least 9000 blocks of contiguous space on the fixed disk.

8. The screen now shows the contents of the partition table and the prompt:

```
Enter the starting user cylinder number [0-n]:
```

Type the starting cylinder number for the *user* partition and press ENTER. The *user* partition must have at least 6000 blocks of contiguous space on the fixed disk.

9. You are prompted:

```
Enter the size of the user partition in K blocks  
(at least 6000 blocks):
```

Respond with the size of the user partition and press ENTER. State the size in 1-K blocks.

10. The **hdinit** program displays the new partition table and the prompt:

```
Is the information correct? (y,n):
```

If all the information is correct, type 'Y' and press ENTER. The **hdinit** program saves the table to the fixed disk. If the table is not correct, type 'N' and press ENTER to enter in information again.

You can now continue with the regular installation. Proceed to Step 7 of "Initializing Your Fixed Disk."

Appendix B

Altering Swap Partition Size

When you install your XENIX system for the first time, your root partition will, by default, occupy 9000 1K blocks. Of those 9000 blocks, XENIX allocates 9 blocks for the boot software and 5991 for the root file system. The remaining 3000 blocks are devoted to the system's swap area. When you install your XENIX system by running the `hdinit` shell script program, the last 3000 blocks (swap size known by the kernel) of the root partition are reserved for the swap area.

There is a one-to-one relationship between the size of the swap area and the maximum amount of memory available to any one user process (`MAXMEM`). If you install more than 3 megabytes of RAM to your XENIX system, you create a situation where `MAXMEM` exceeds the size of the swap area. When you boot your system, you will see this message:

```
Maxmem was reduced based on the size of the swap area.  
Refer to the system documentation for information on  
the relationship between memory size and swap size.
```

You can refer to **messages (M)** in the *Reference Manual Volume I* for further explanation of this message.

When this message appears, XENIX can only allocate 3000 Kbytes to a process even if you have installed more memory. In this way XENIX enforces the one-to-one relationship between swap size and `MAXMEM`.

To increase the swap size from the 3000 block default, follow these three steps:

1. Back up any files you wish to save from your primary fixed disk.
2. Reinstall your system using the original installation set and specify the new swap area size.
3. Restore the files you backed up in Step 1.

If you compile your own kernel, or use installable device drivers and you're not using the default swap area size, you'll have to run the `chswap(M)` utility on those kernels. See the documentation on `chswap(M)` for more information.

